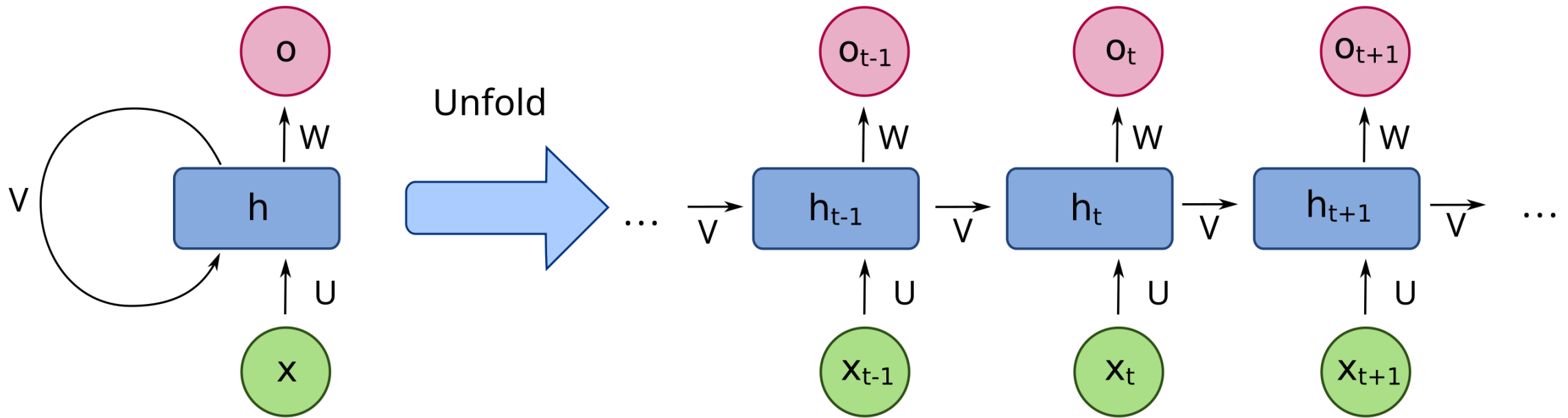




RNN

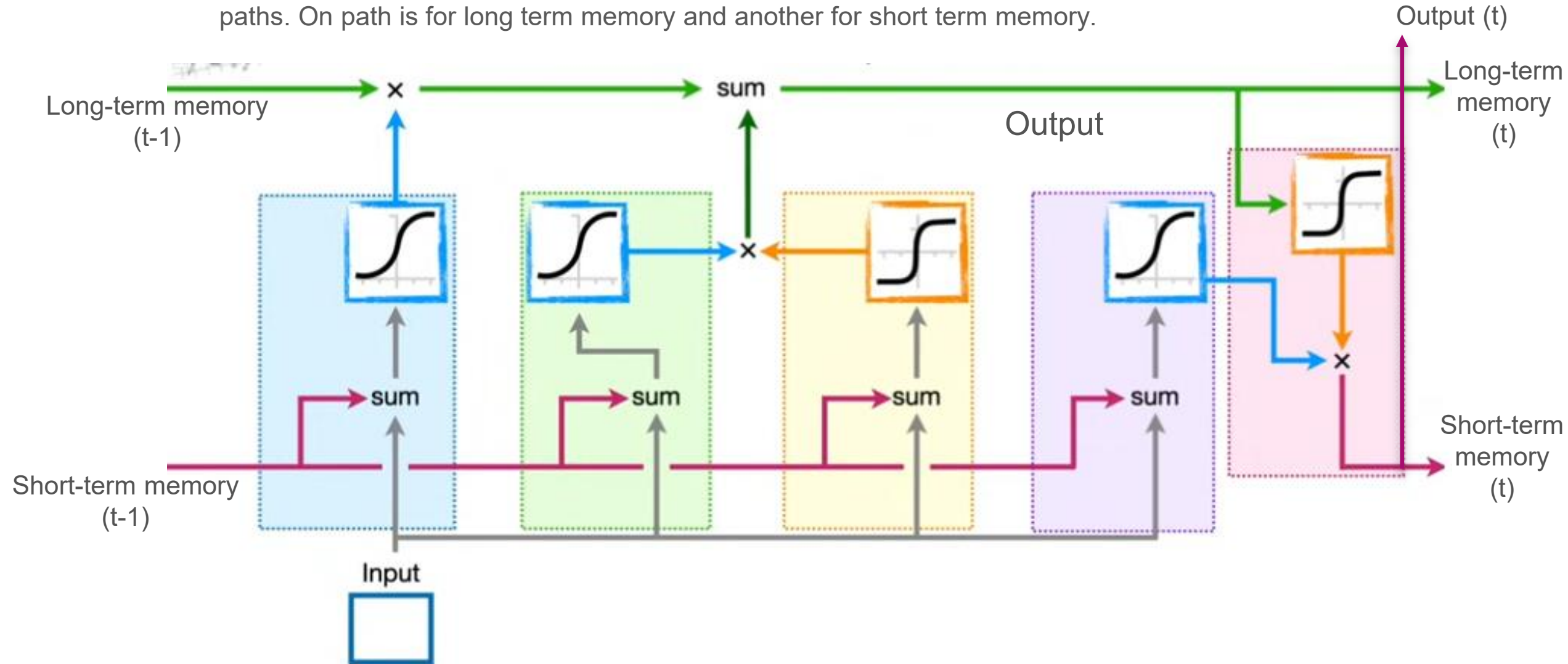
- Recurrent Neural Networks
- LSTM

RNN – See Notebook



LSTM – Long Short Term Memory

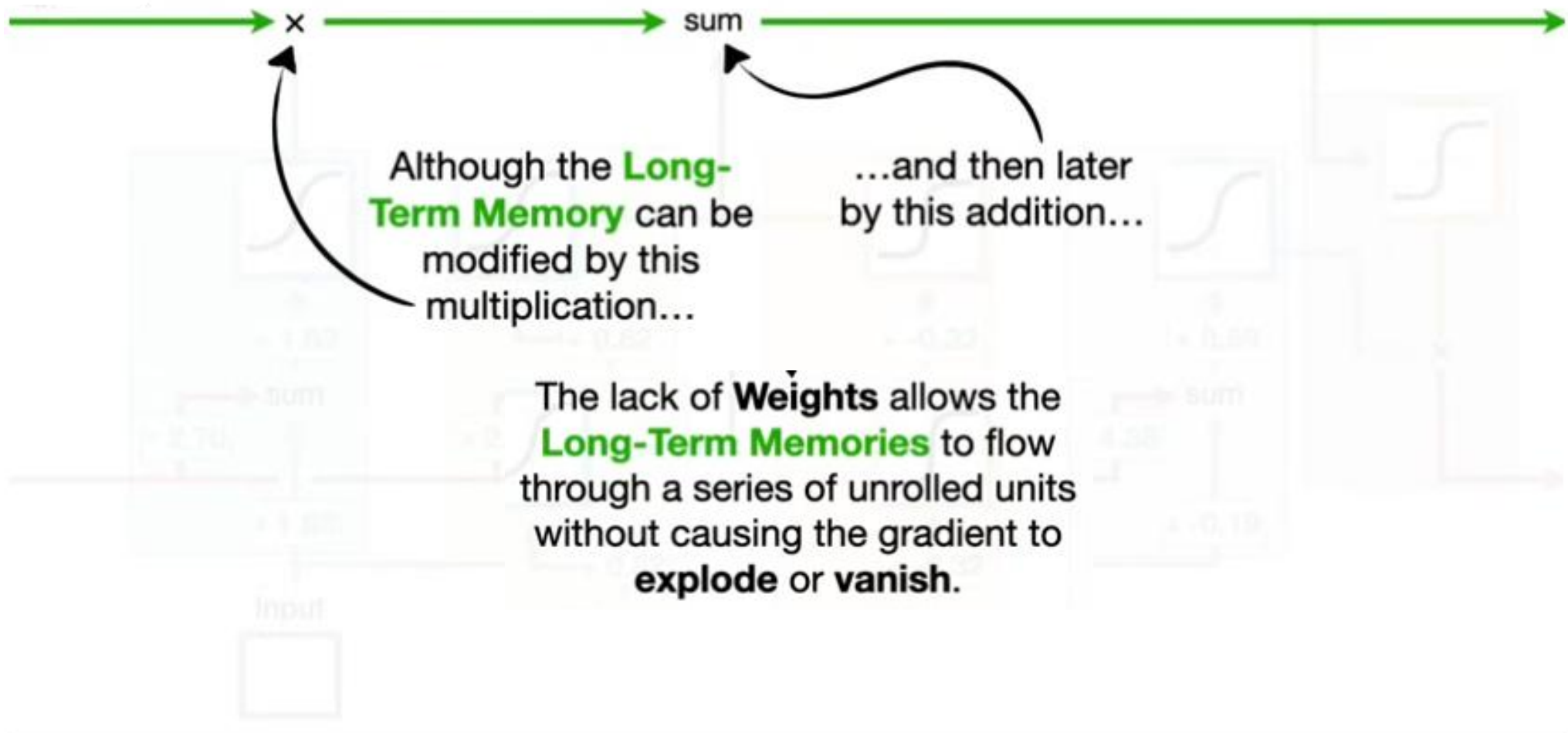
- Main idea:** instead of using the same feedback loop connection for events (data) that happened long ago and events (data) that happened yesterday (e.g., previous time stamp), LSTM uses two distinct paths. One path is for long term memory and another for short term memory.



Long Short-Term Memory uses **Sigmoid Activation Functions...**

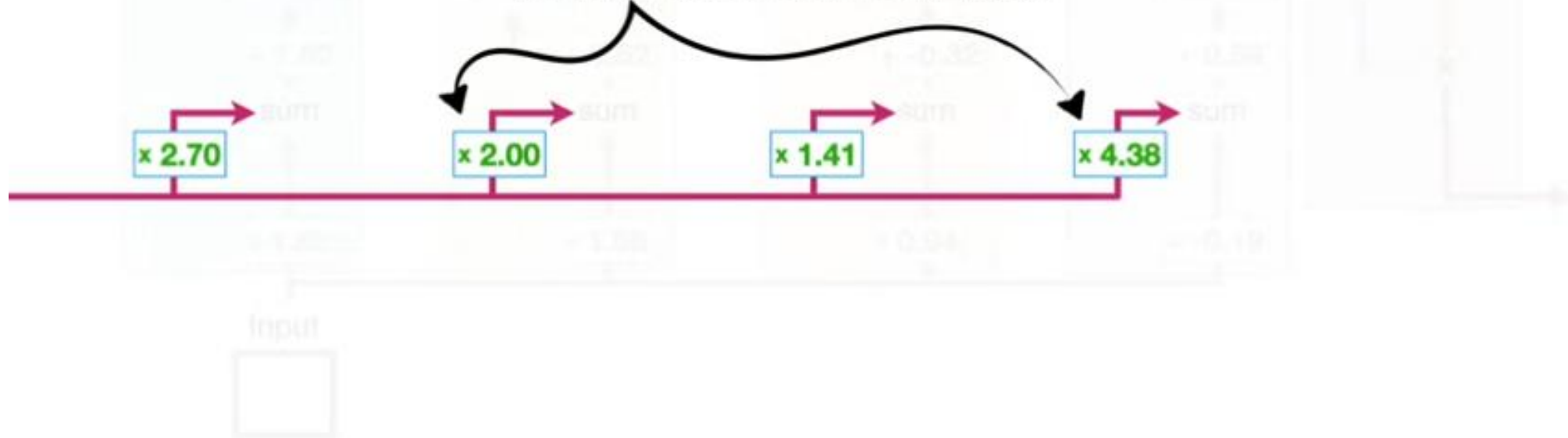


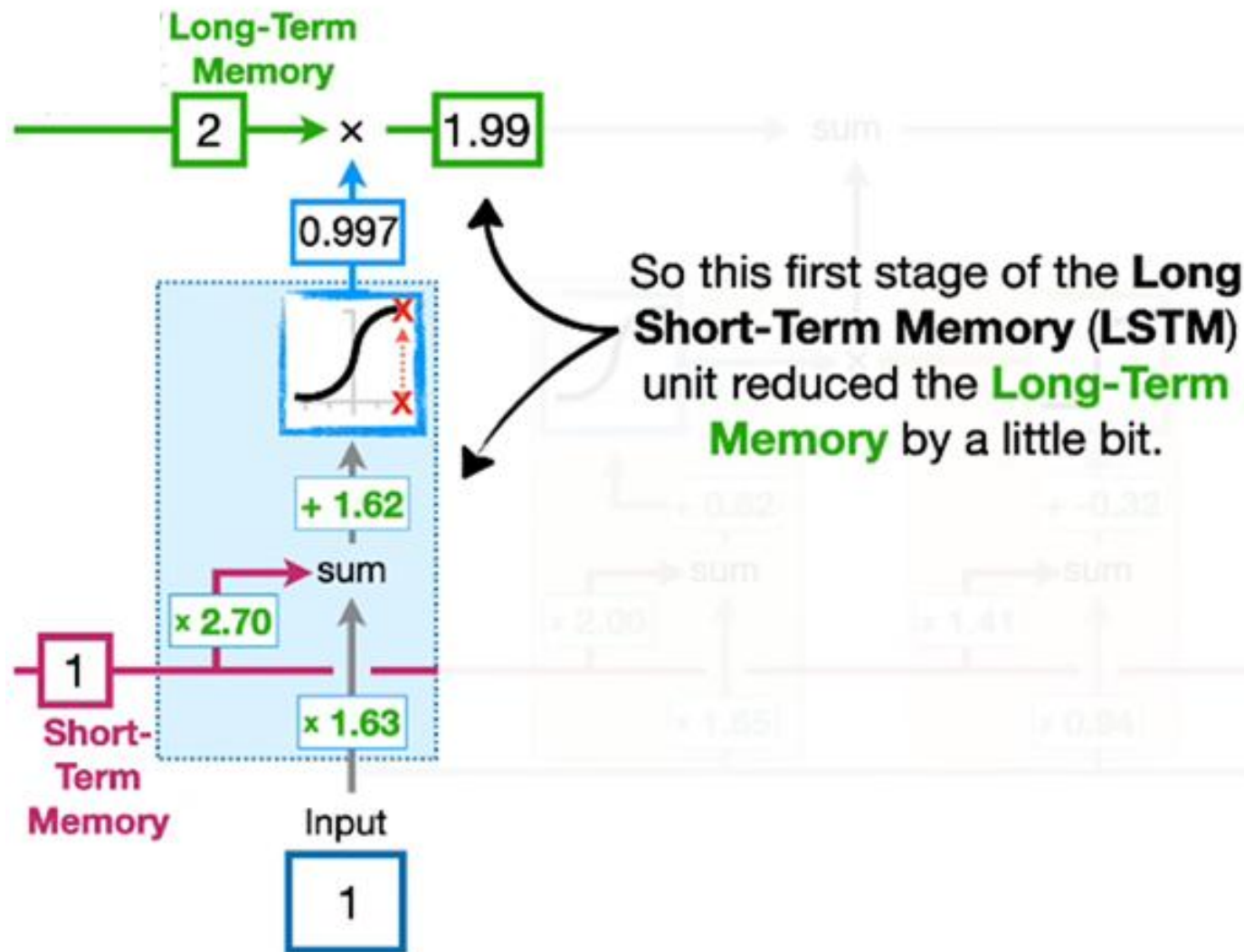
...and **Tanh Activation Functions...**

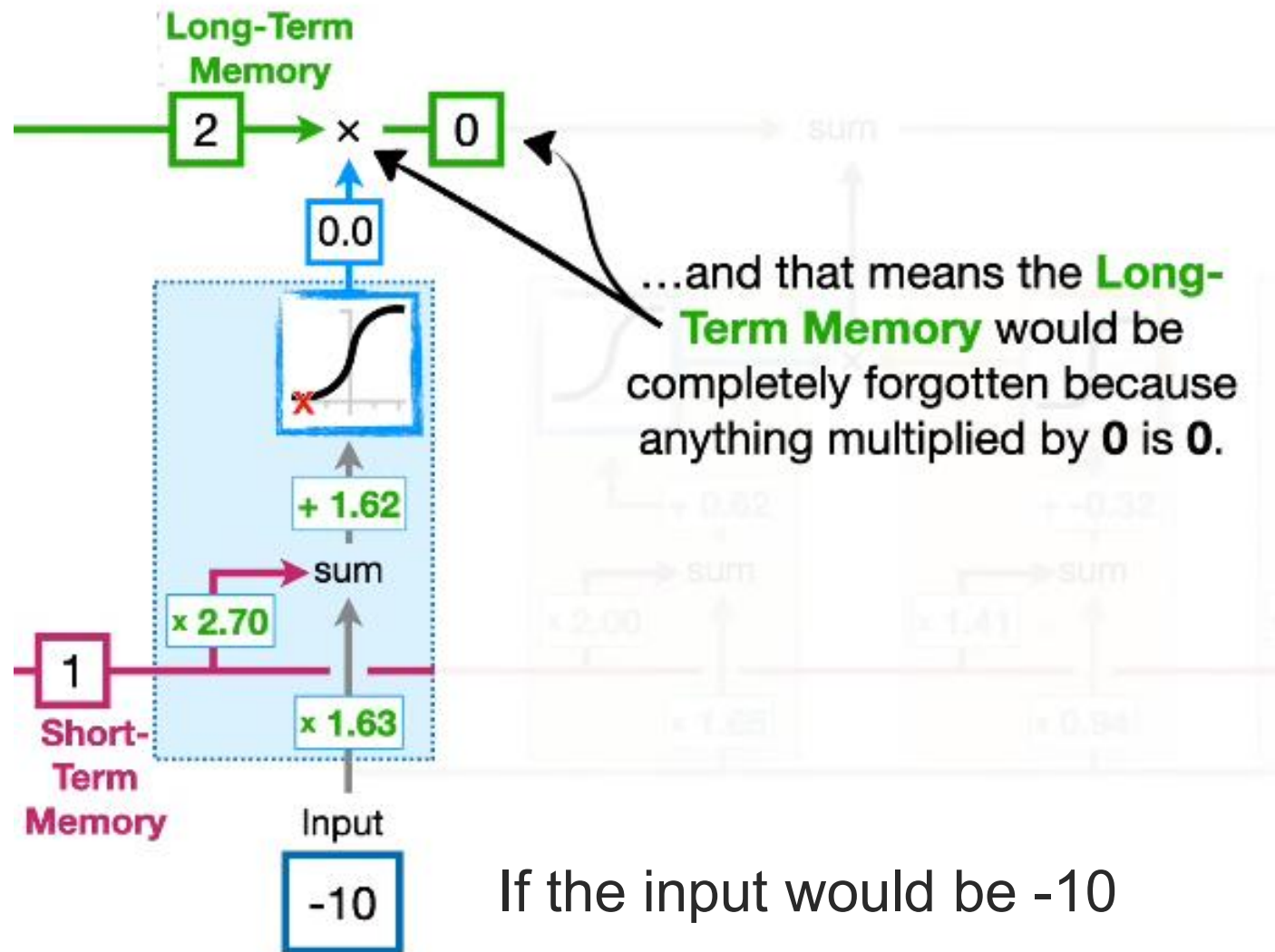




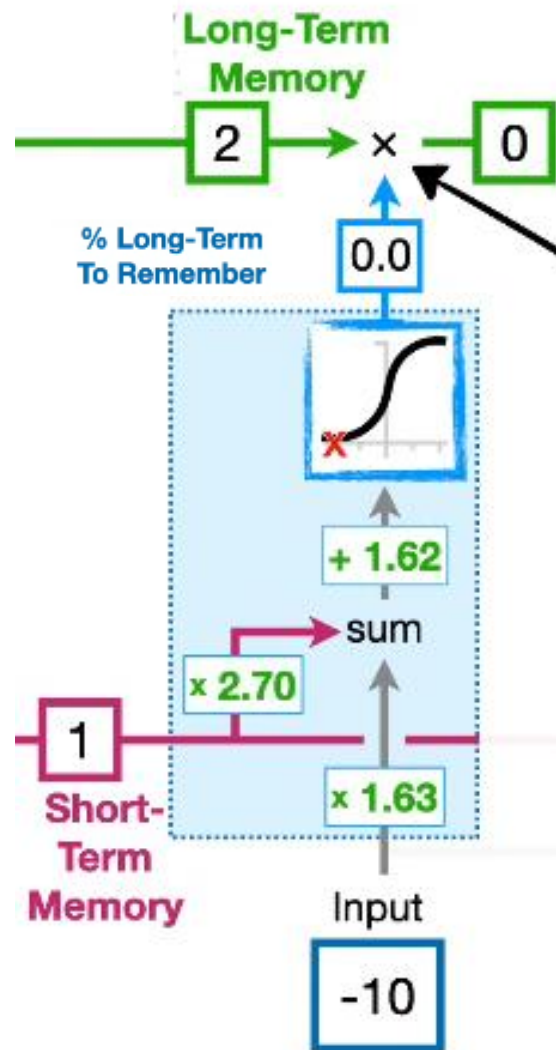
And, as we can see, the **Short-Term Memories** are directly connected to **Weights** that can modify them.







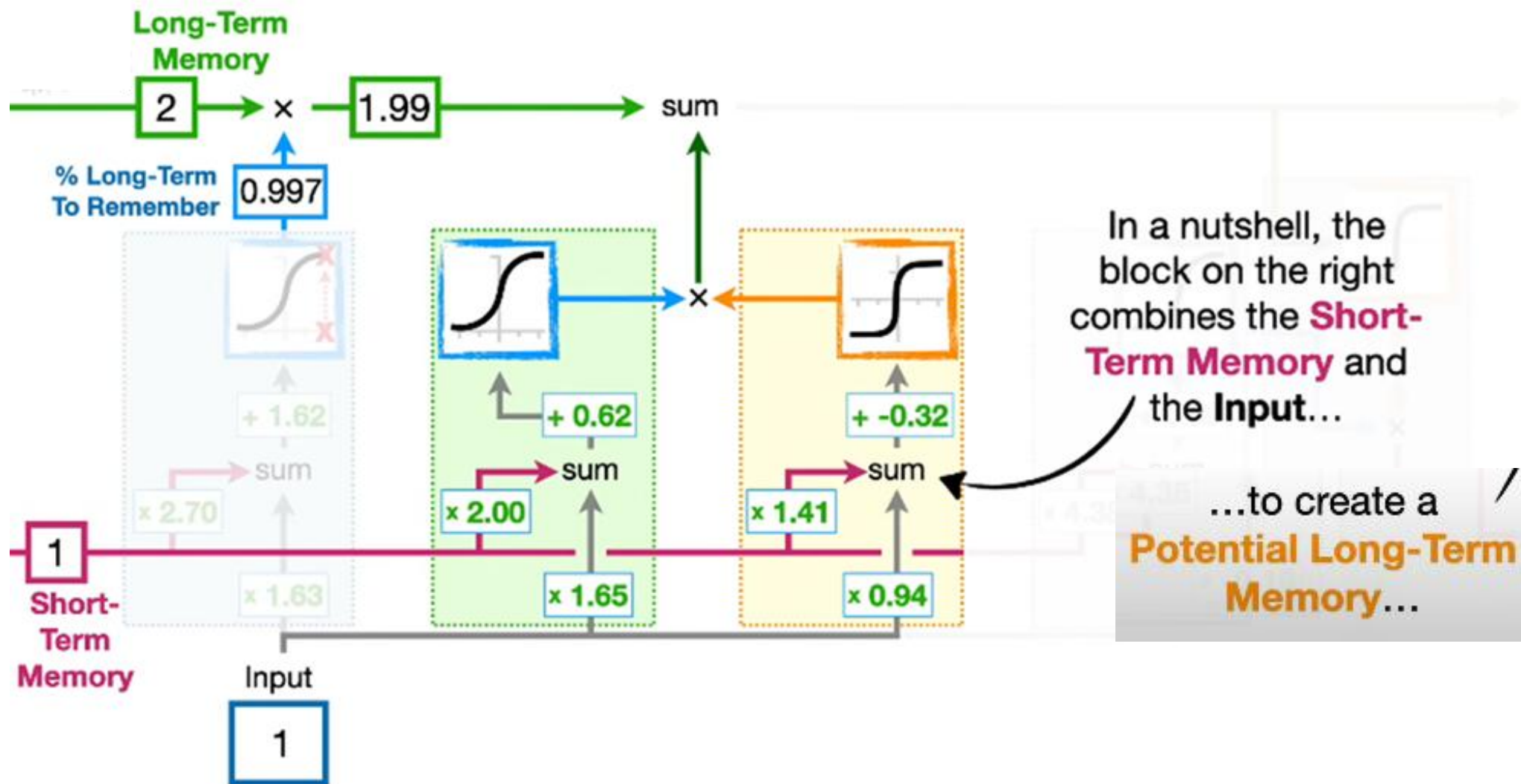
If the input would be -10

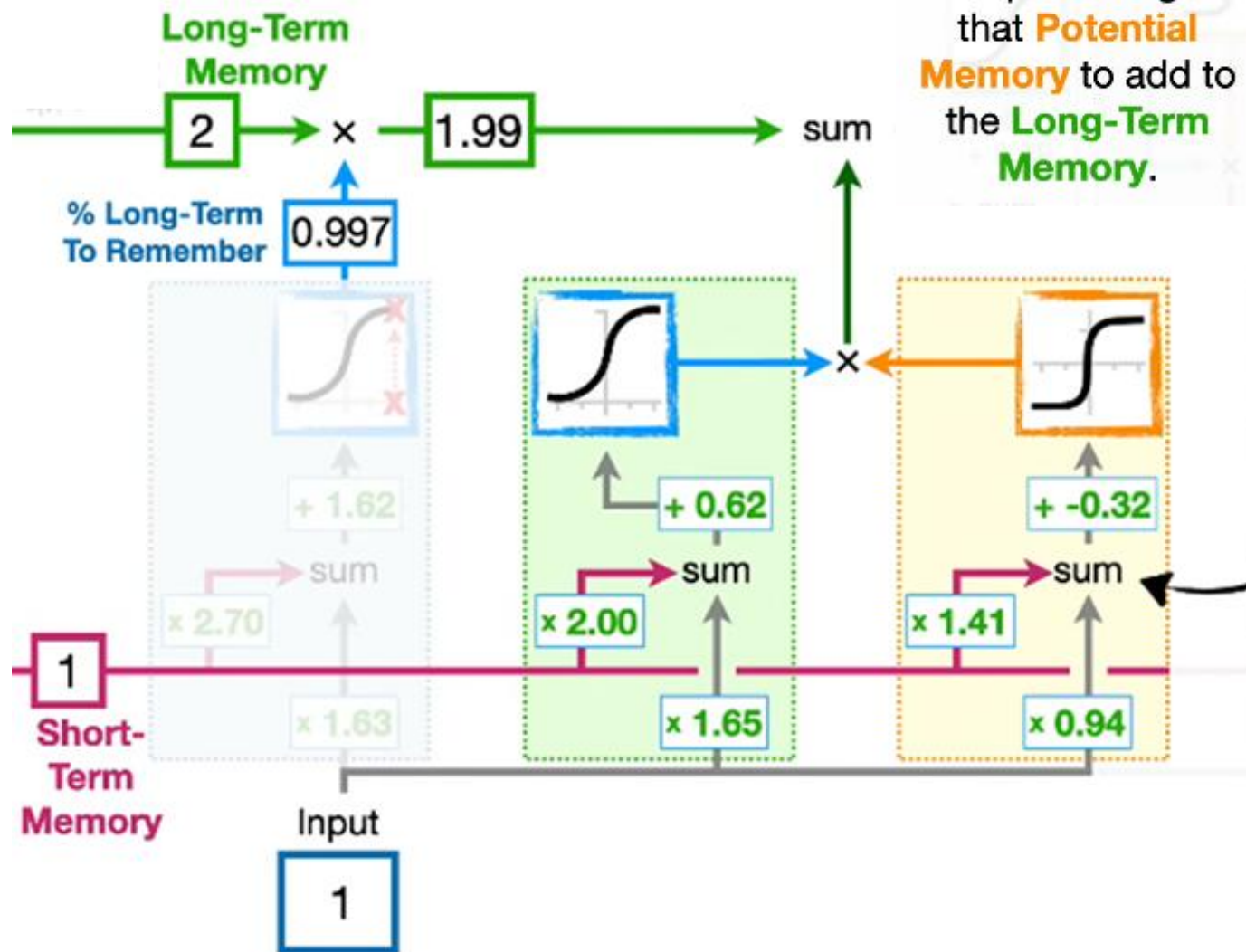


...and that means the **Long-Term Memory** would be completely forgotten because anything multiplied by **0** is **0**.

To summarize, the first stage in a **Long Short-Term Memory** unit determines what percentage of the **Long-Term Memory** is remembered.

Therefore, this part is called the **Forget gate**

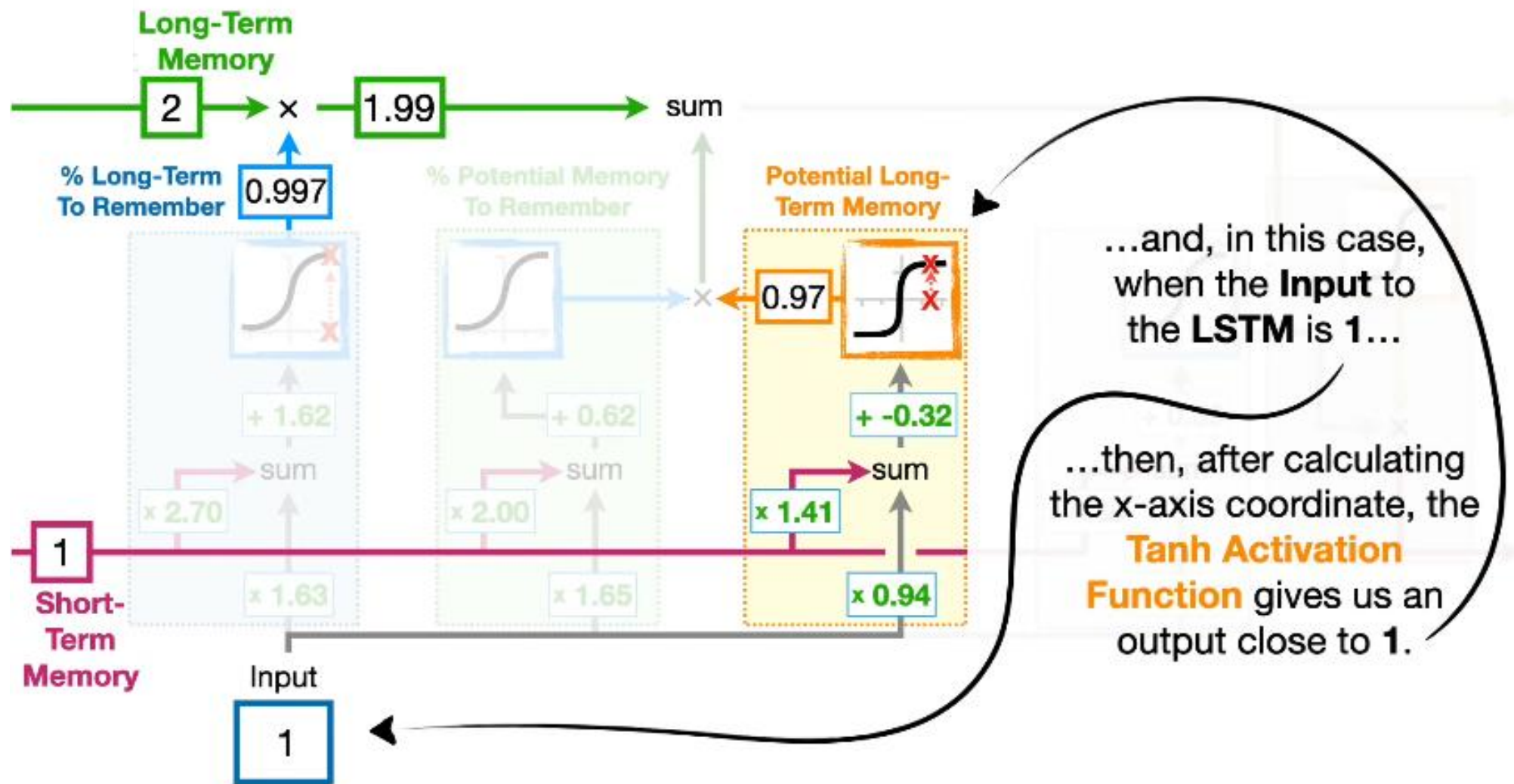


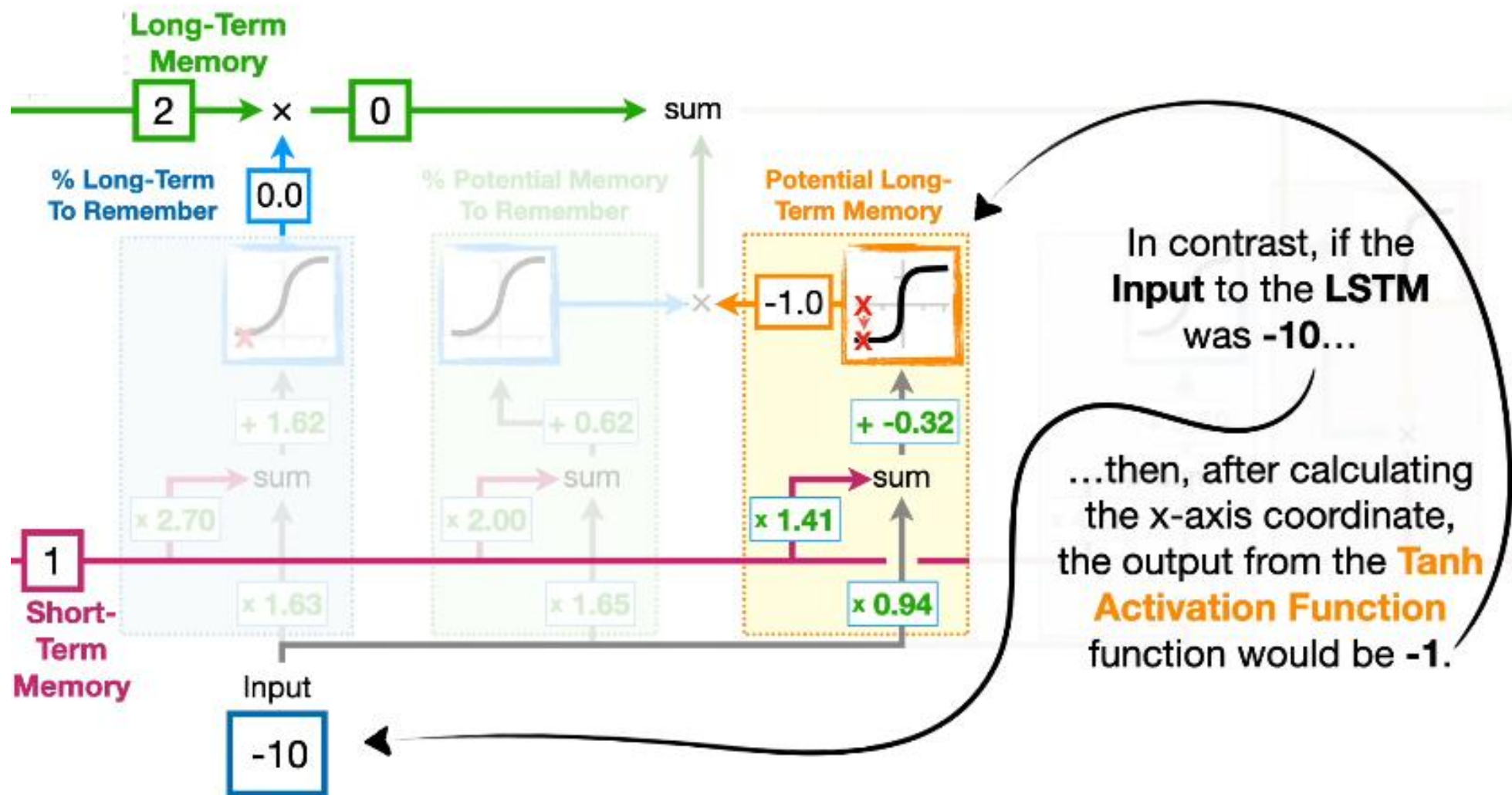


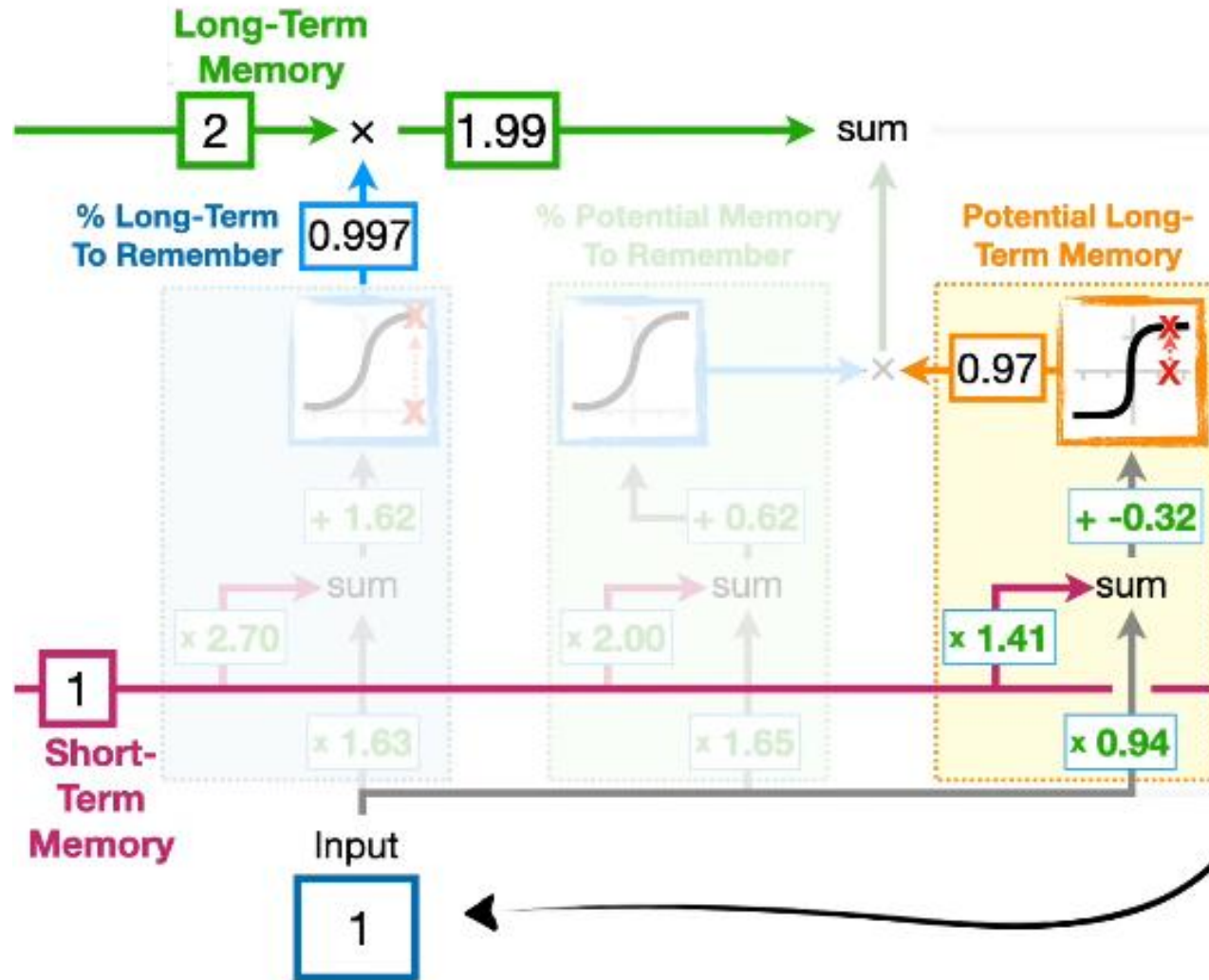
...and the block on the left determines what percentage of that **Potential Memory** to add to the **Long-Term Memory**.

In a nutshell, the block on the right combines the **Short-Term Memory** and the **Input**...

...to create a **Potential Long-Term Memory**...

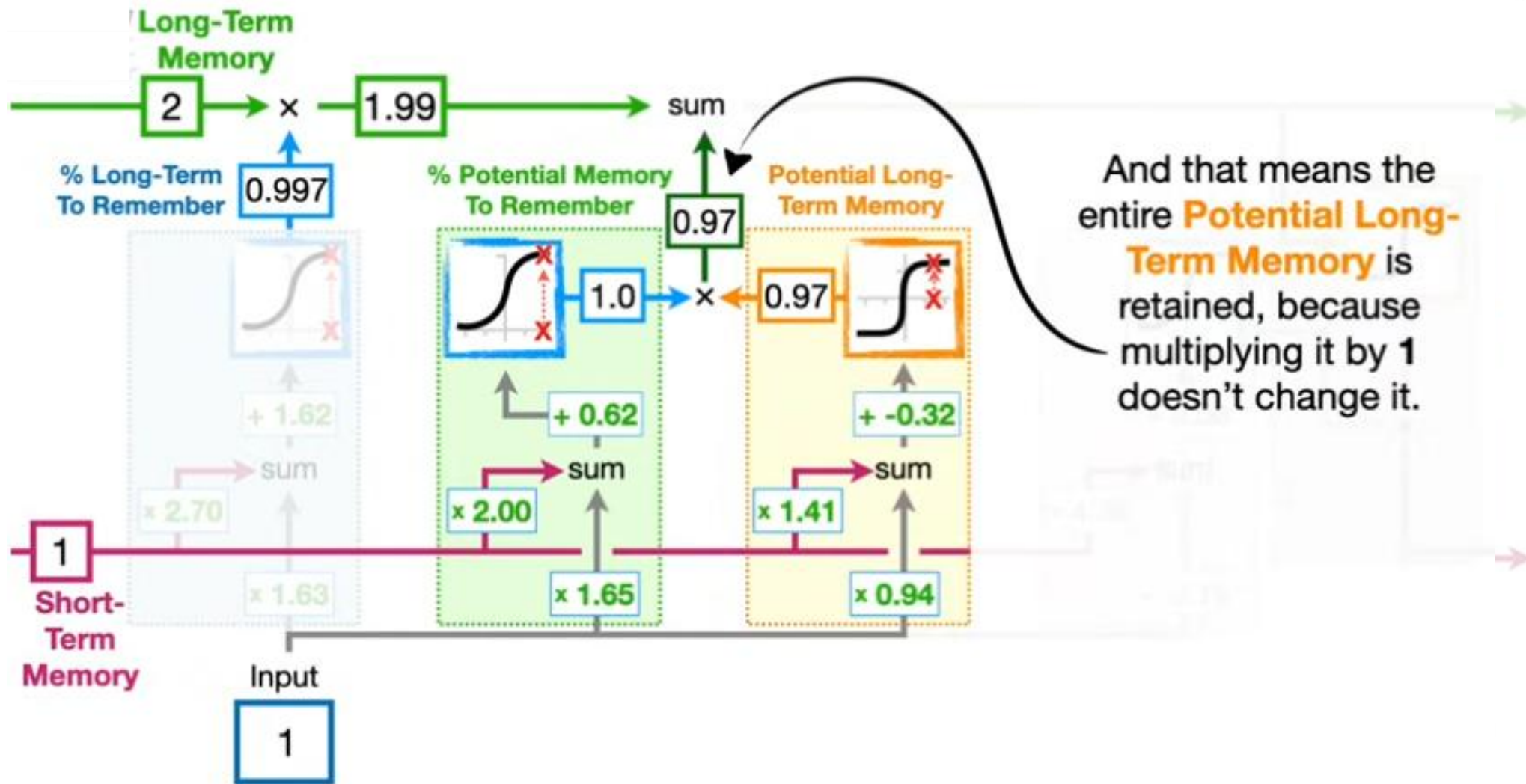


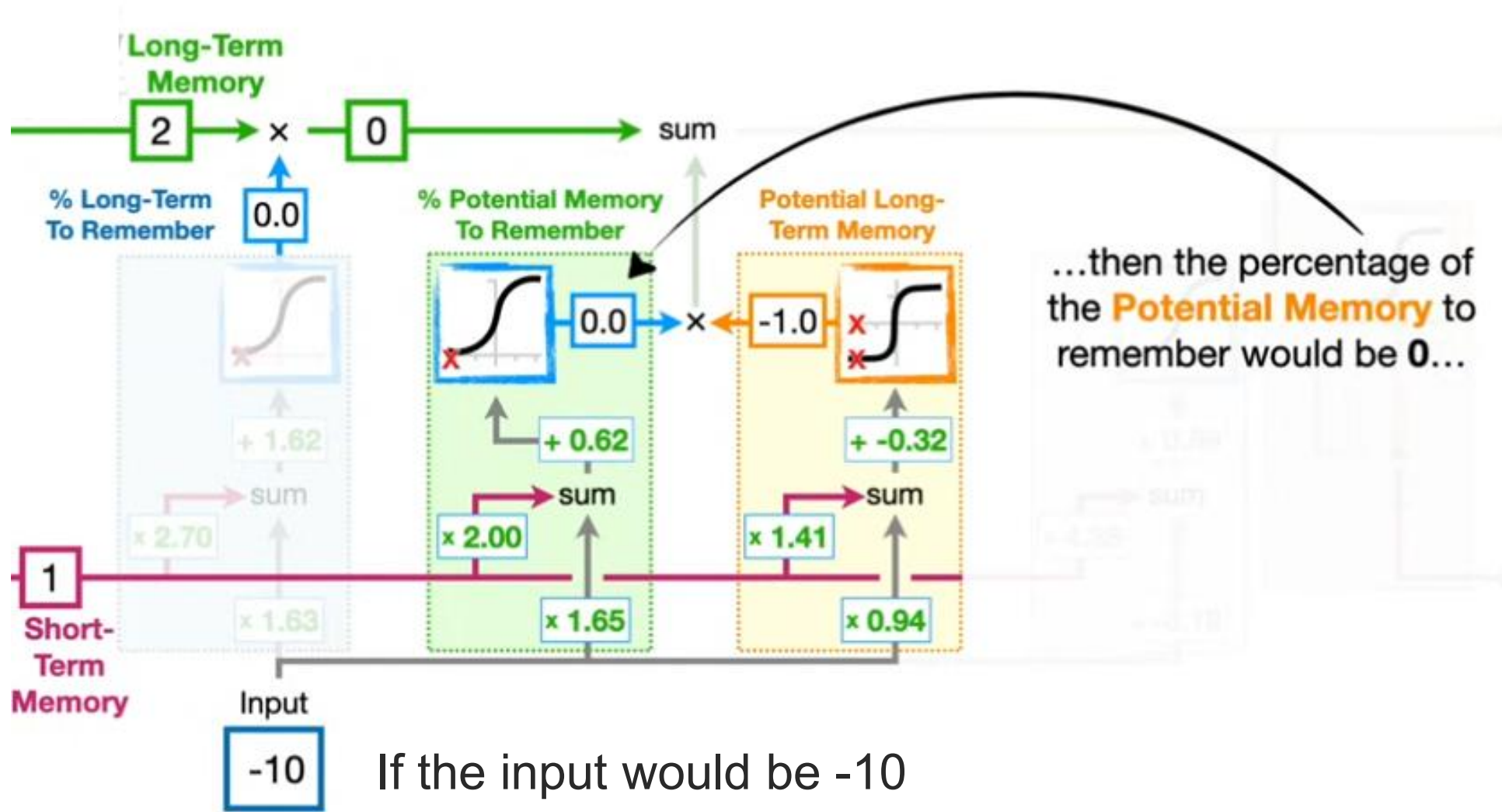




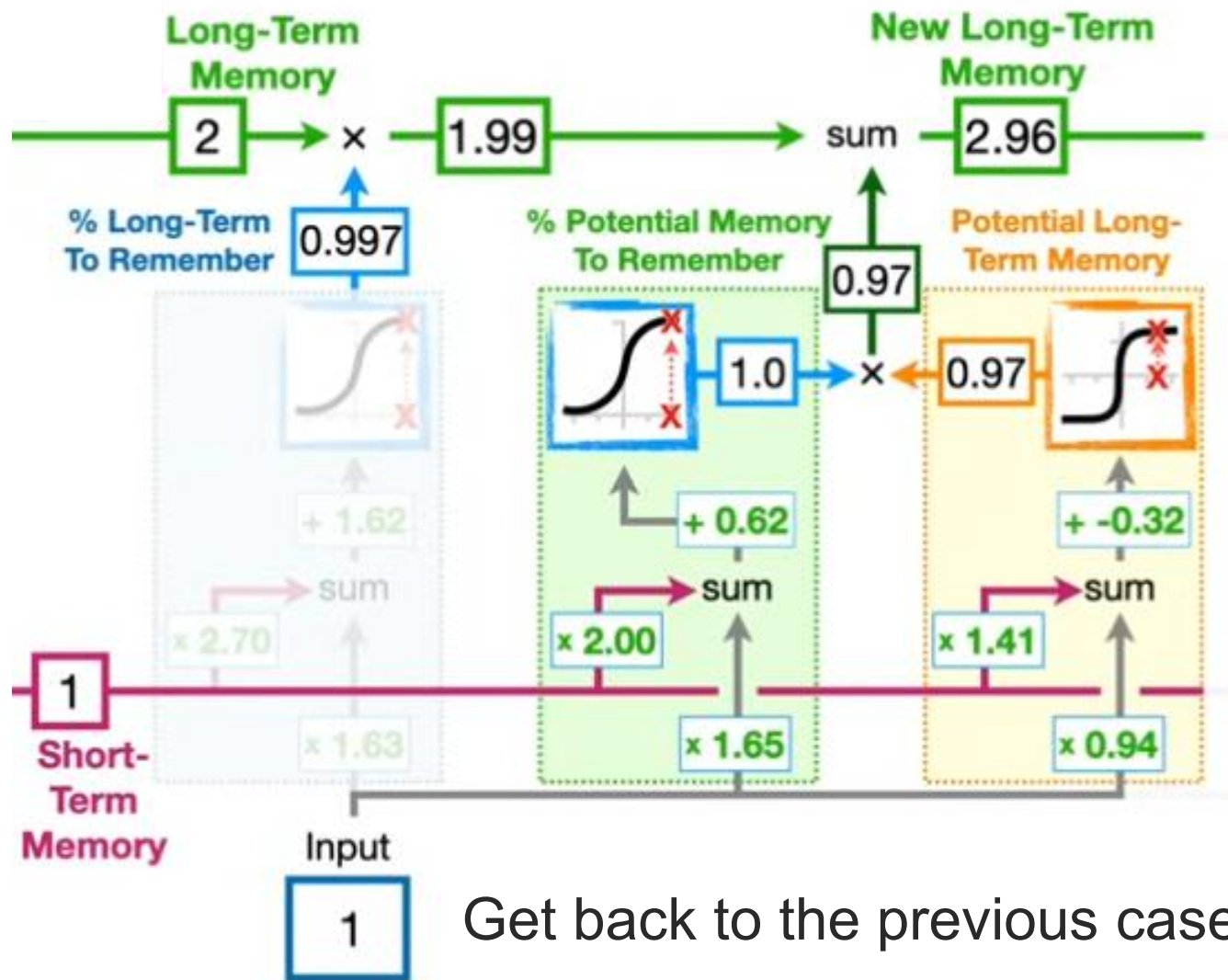
Get back to the previous case

Now the **LSTM** has to decide how much of this **Potential Memory** to save...





If the input would be -10



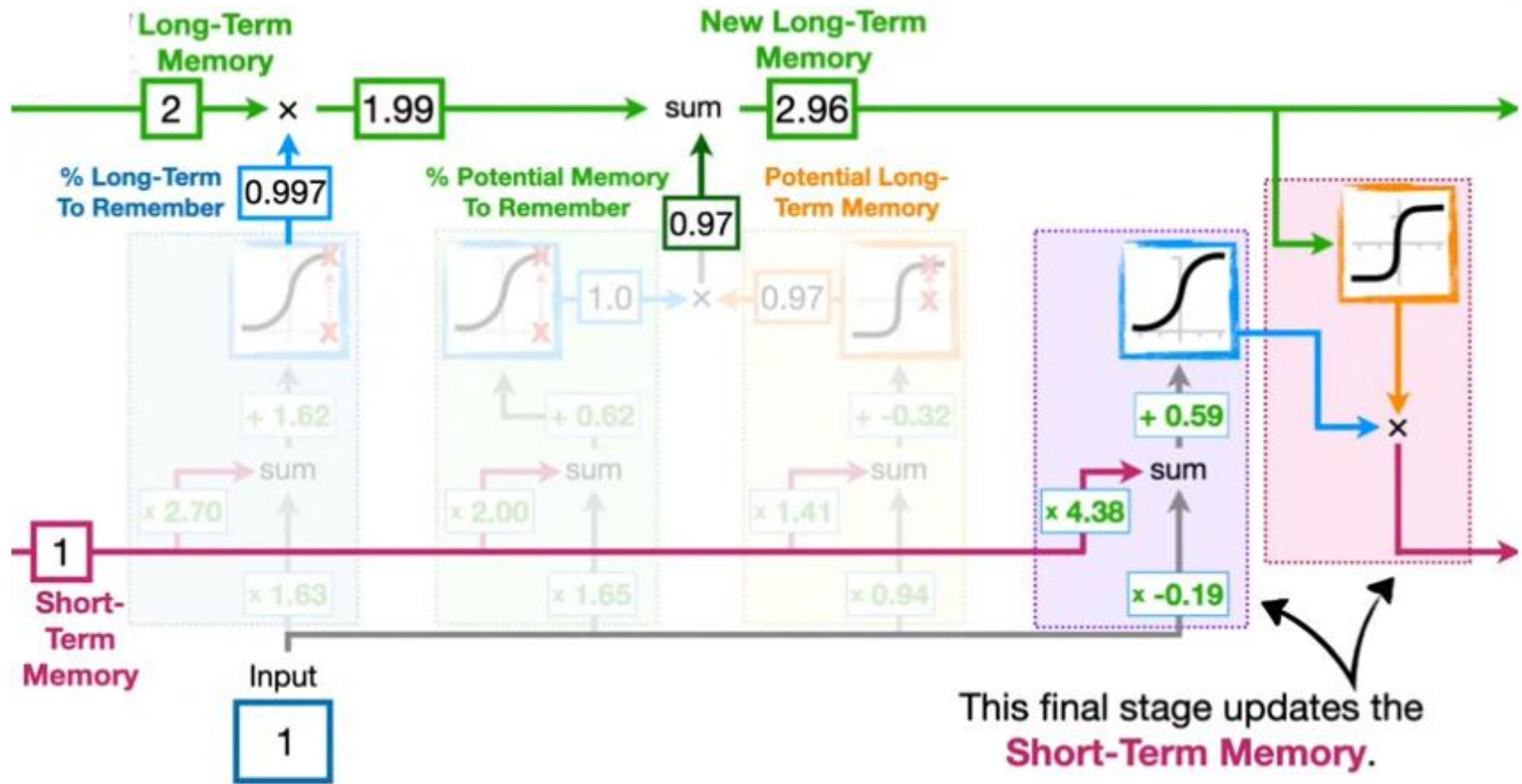
Get back to the previous case

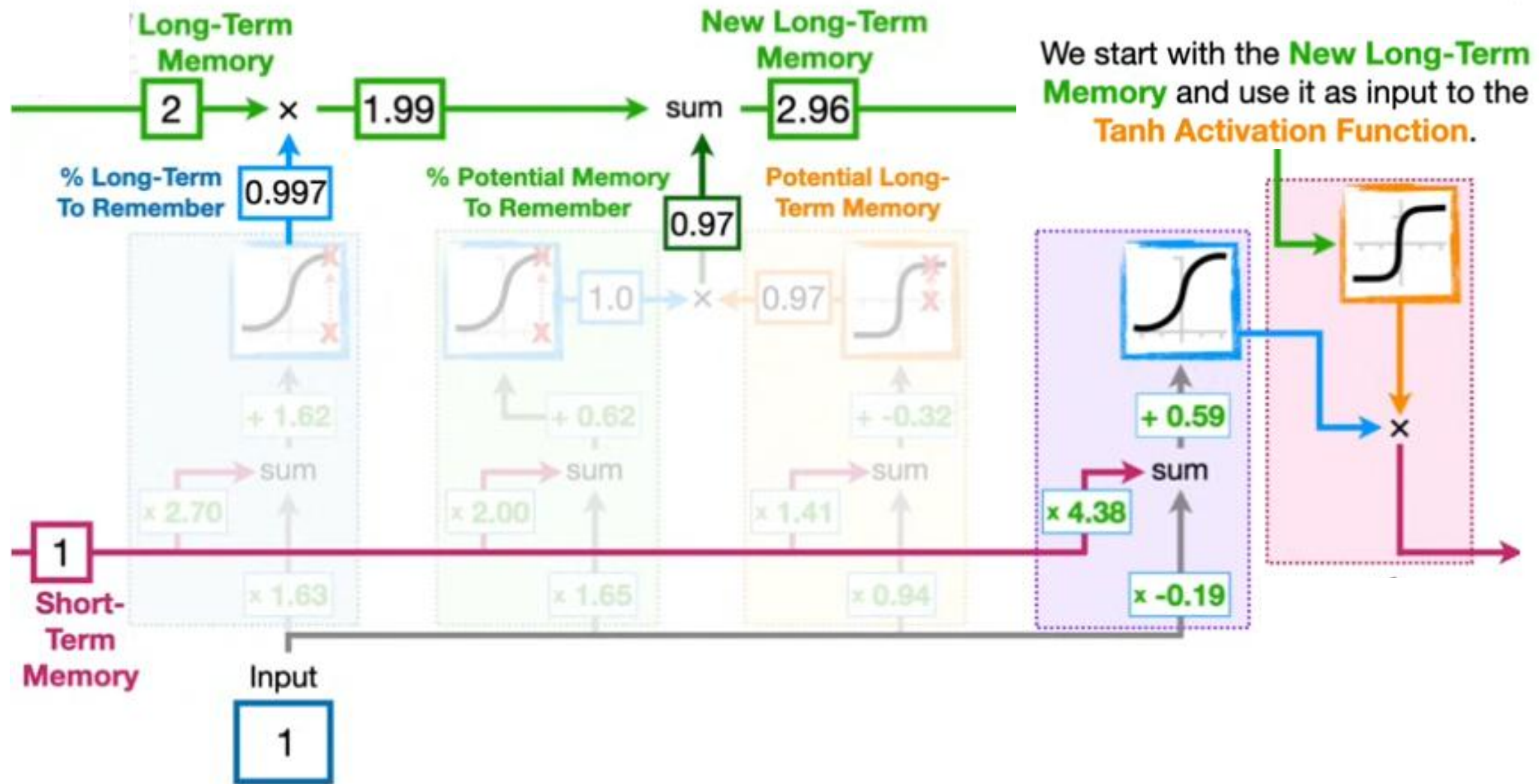
...we add **0.97** to the existing **Long-Term Memory**...

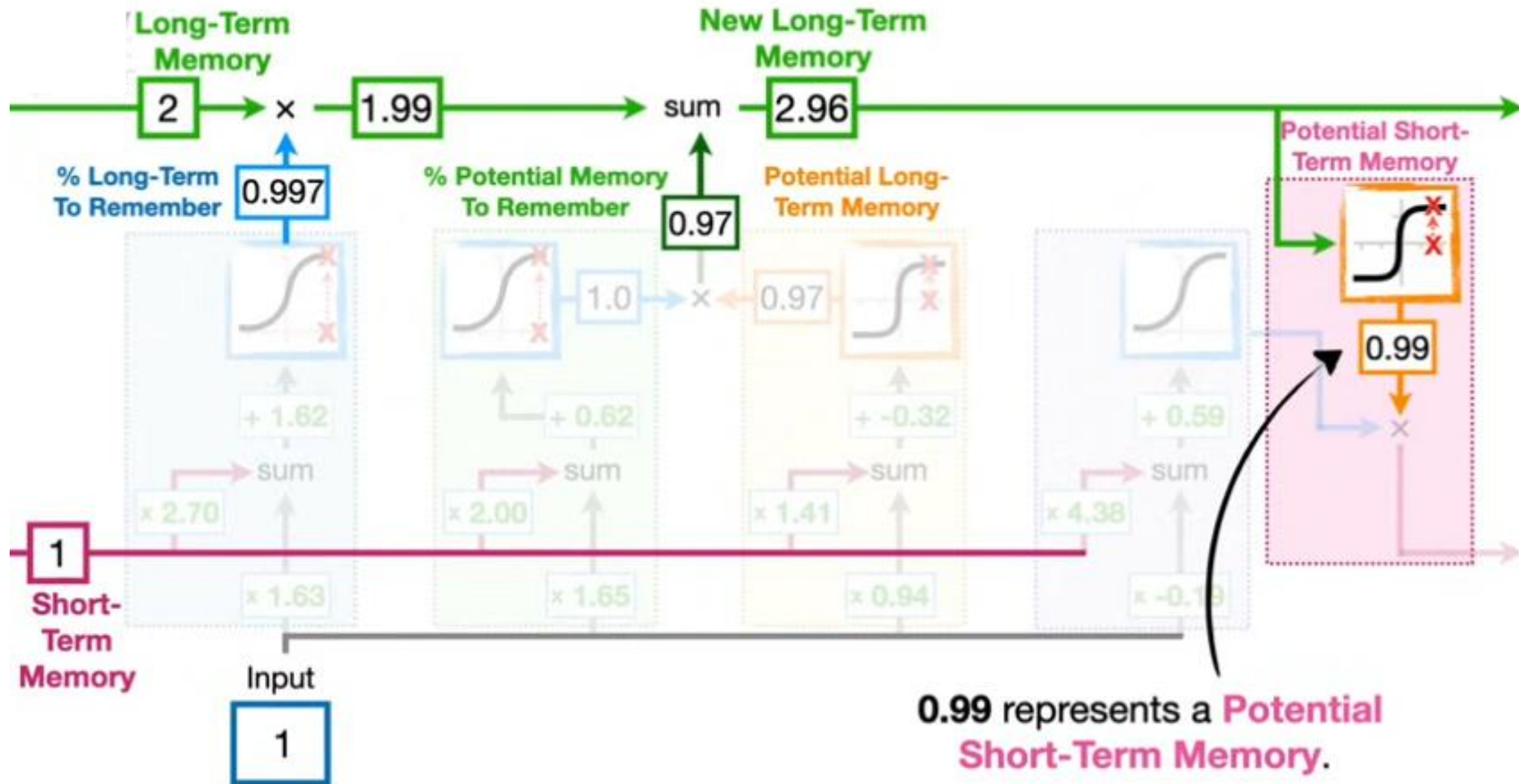
...and we get a new **Long-Term Memory**, **2.96**.

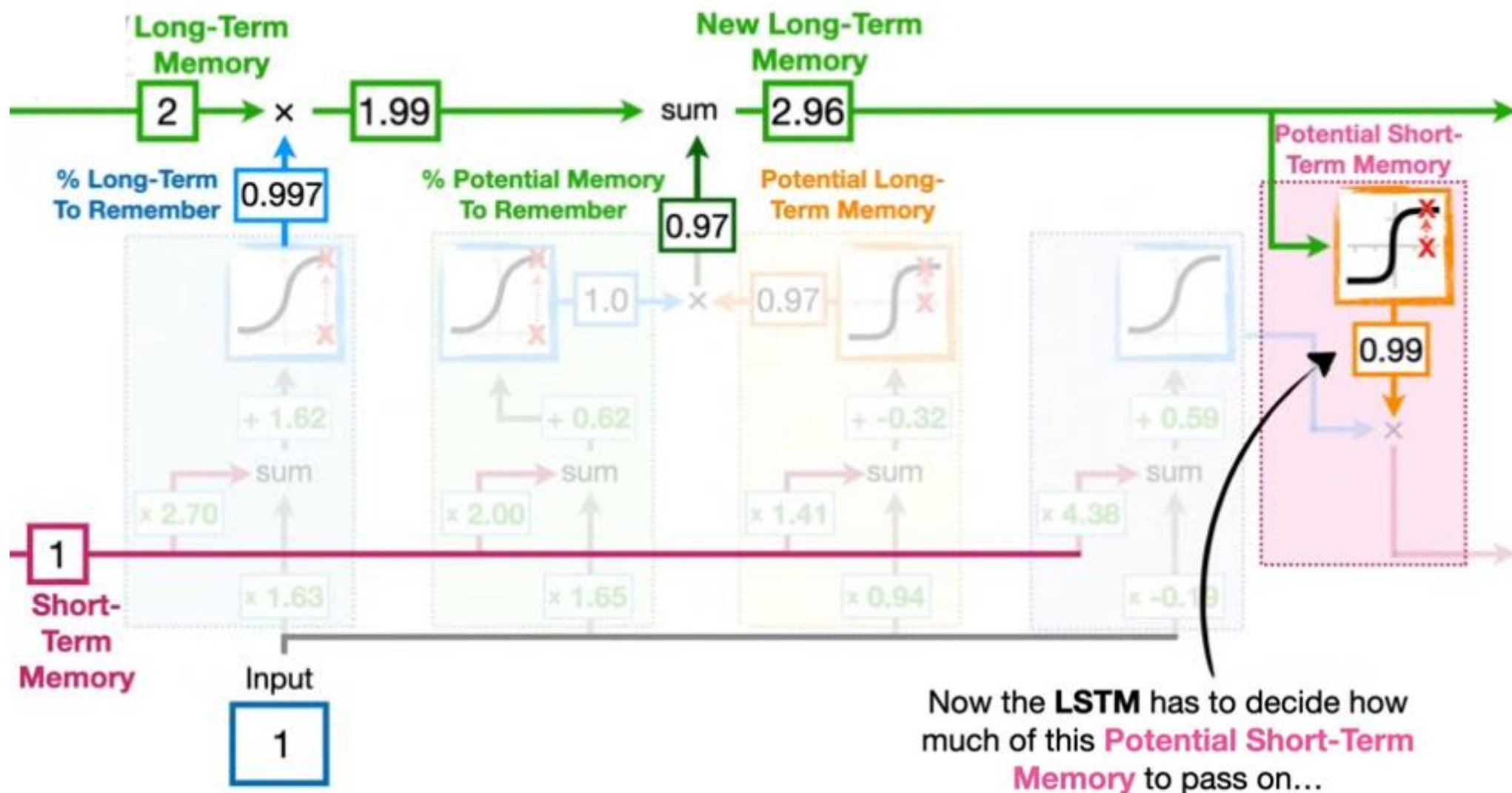
Even though this part of the **Long Short-Term Memory** unit determines how we should update the **Long-Term Memory**...

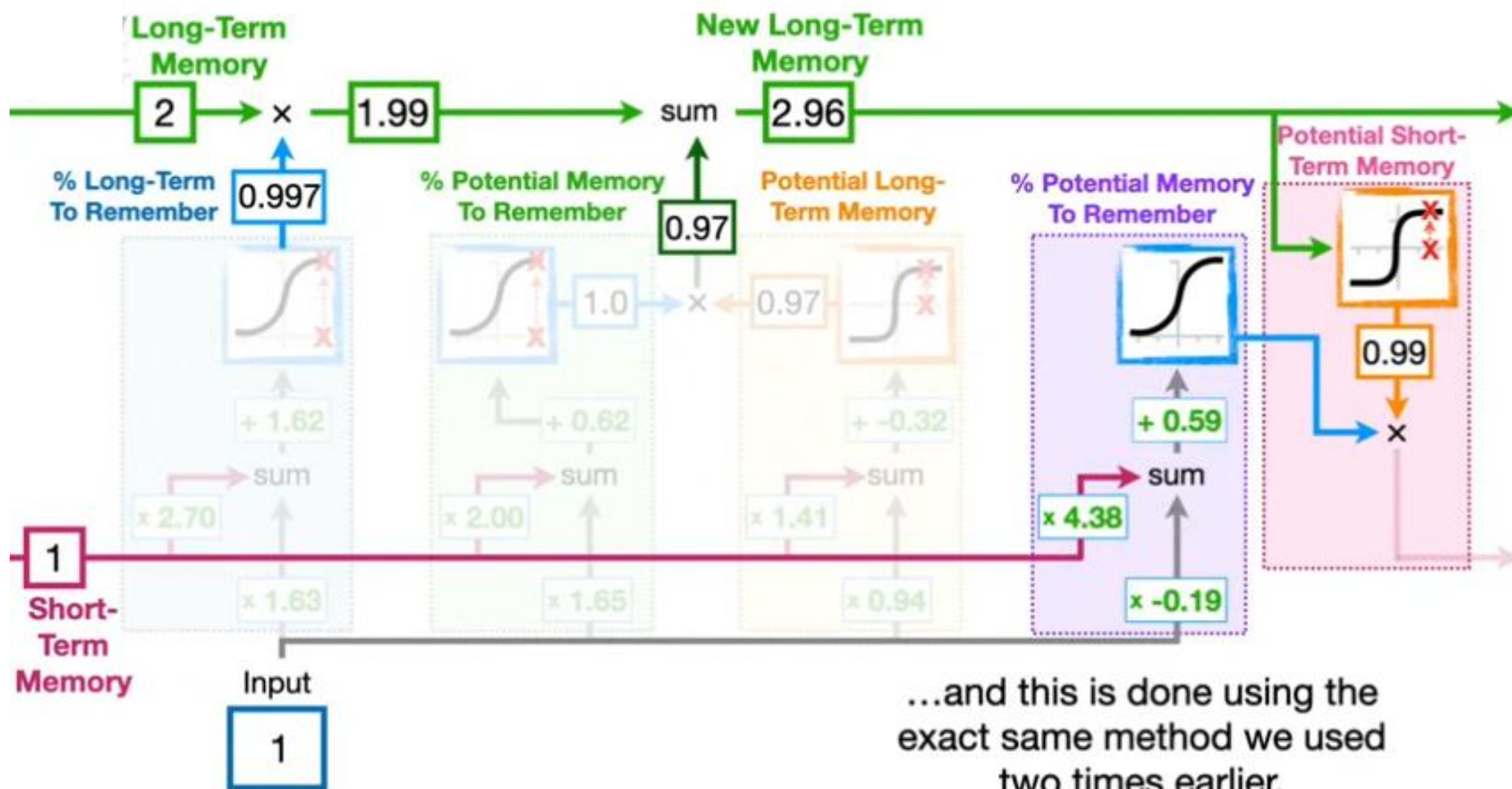
...it is usually called the **Input Gate**.



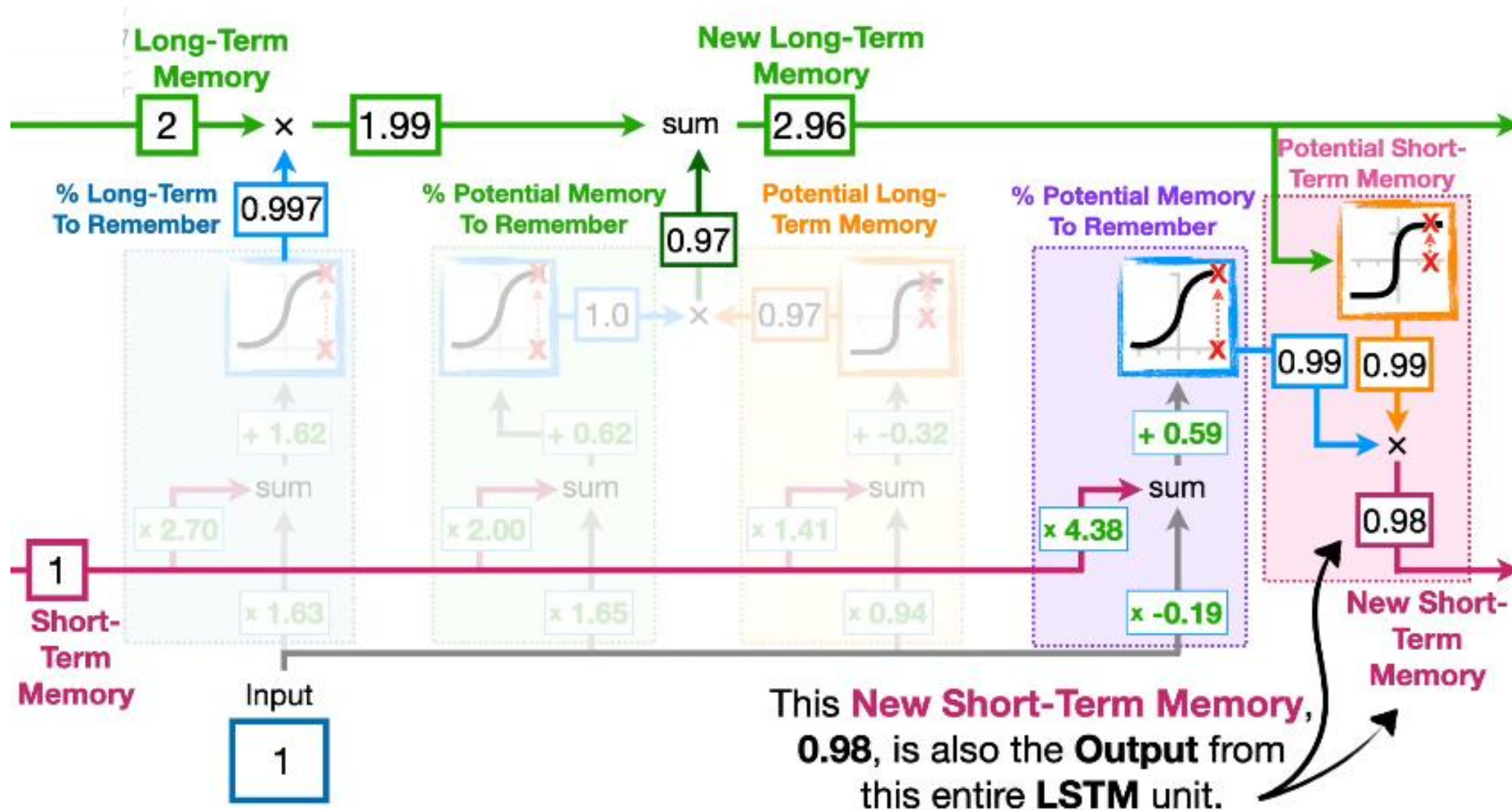






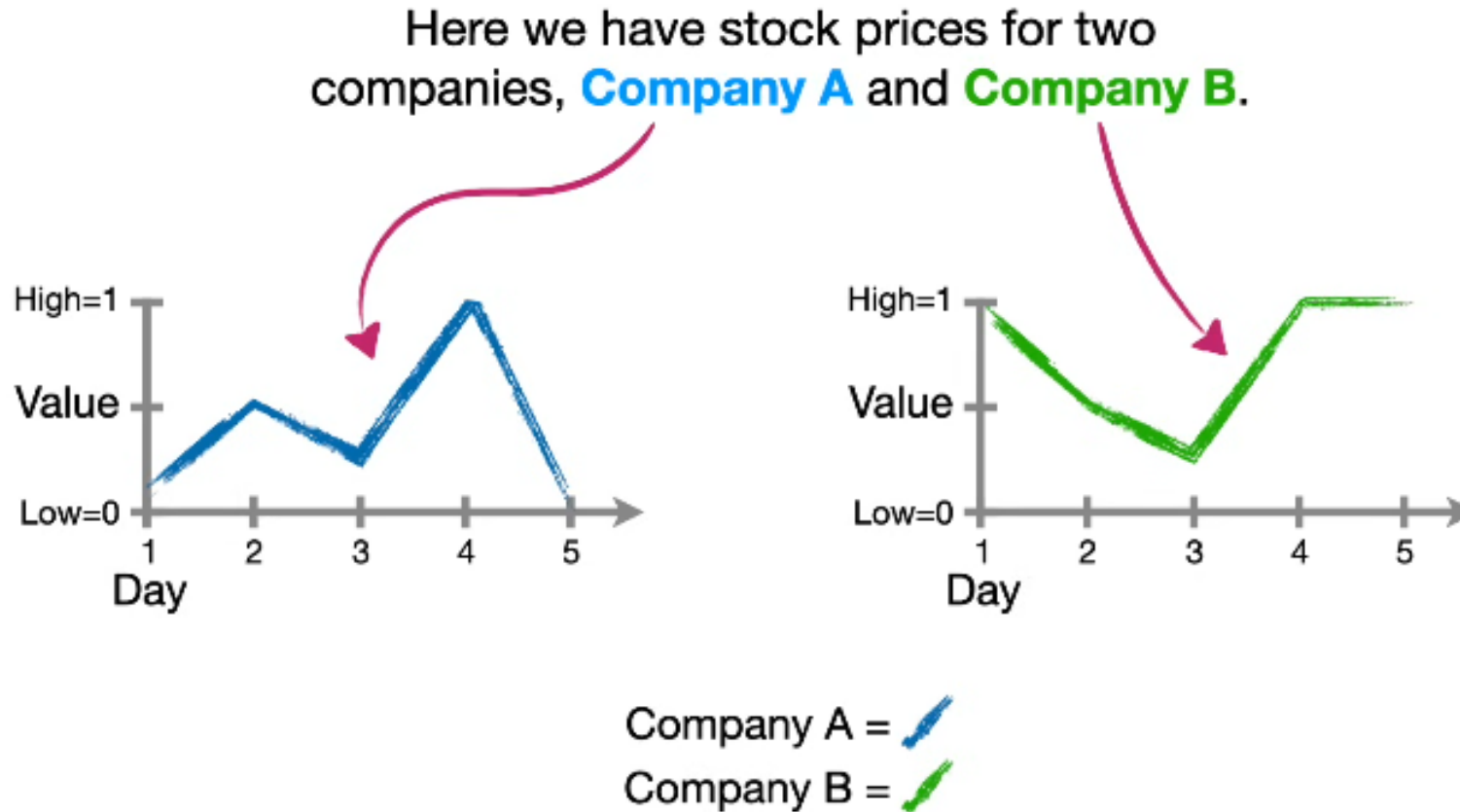


...and this is done using the exact same method we used two times earlier.

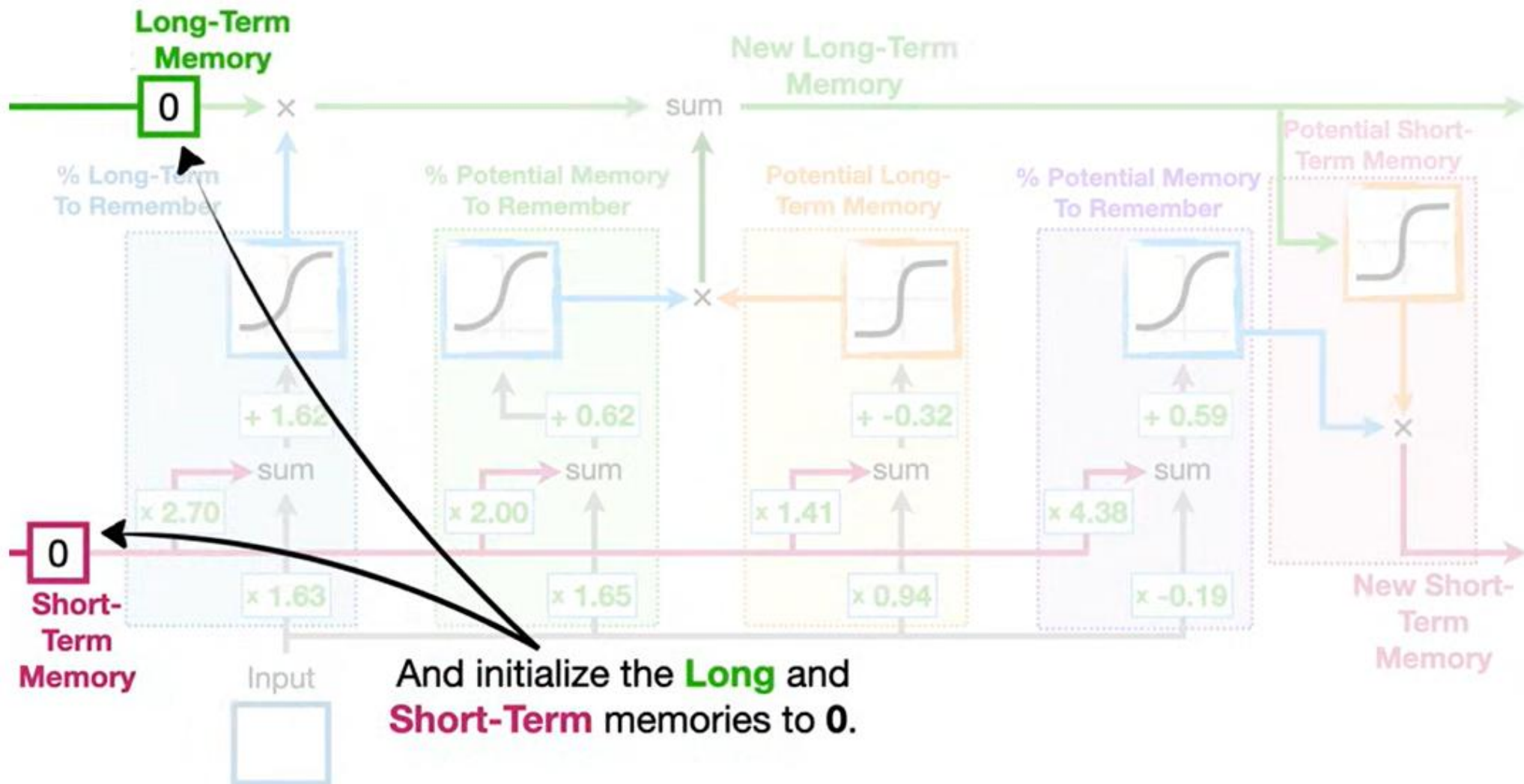


This part is called the **Output gate**

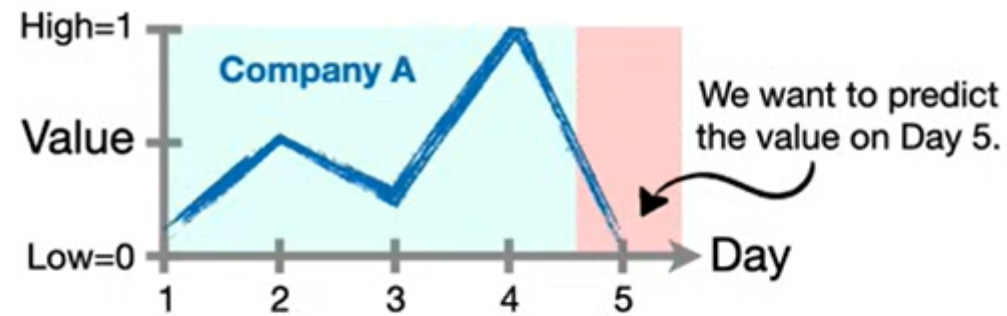
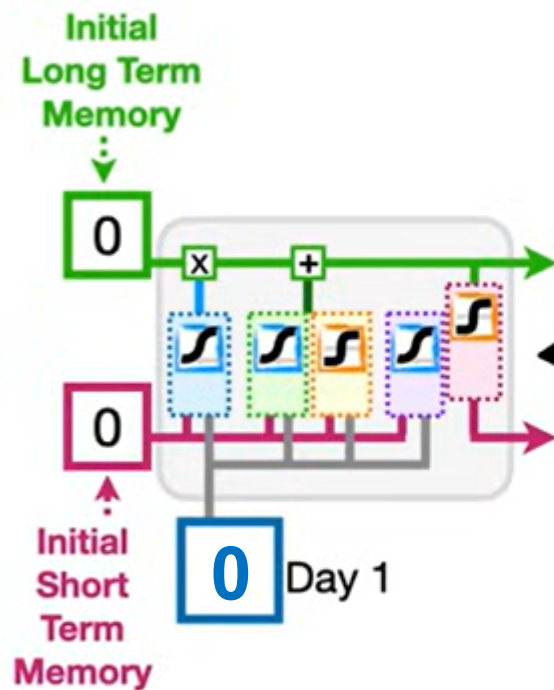
Let's see an example



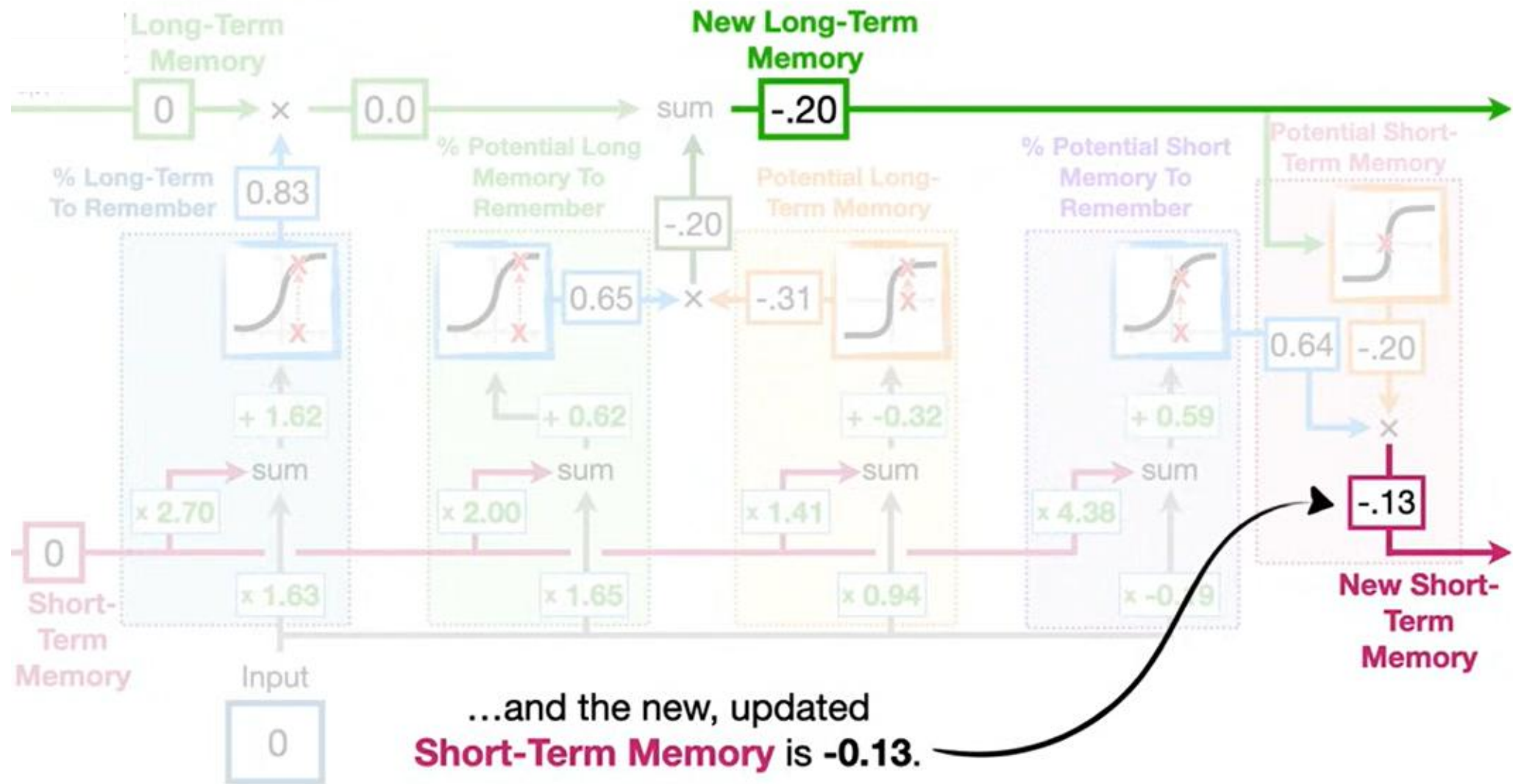
The only difference is at day 1 and day 5... we want the LSTM to remember what happened on day 1 to predict what will happen in day 5

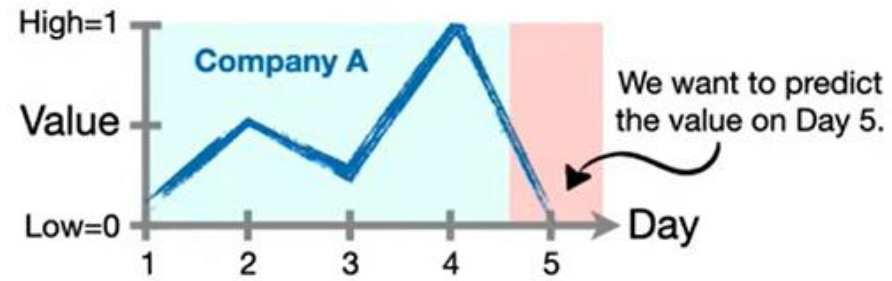
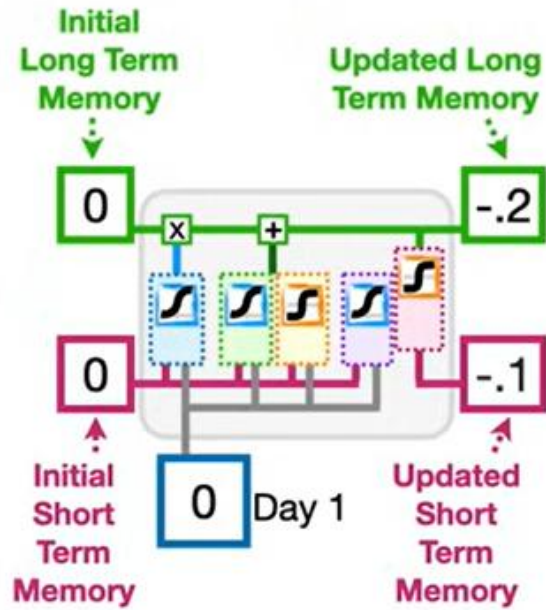


And initialize the **Long** and **Short-Term** memories to 0.

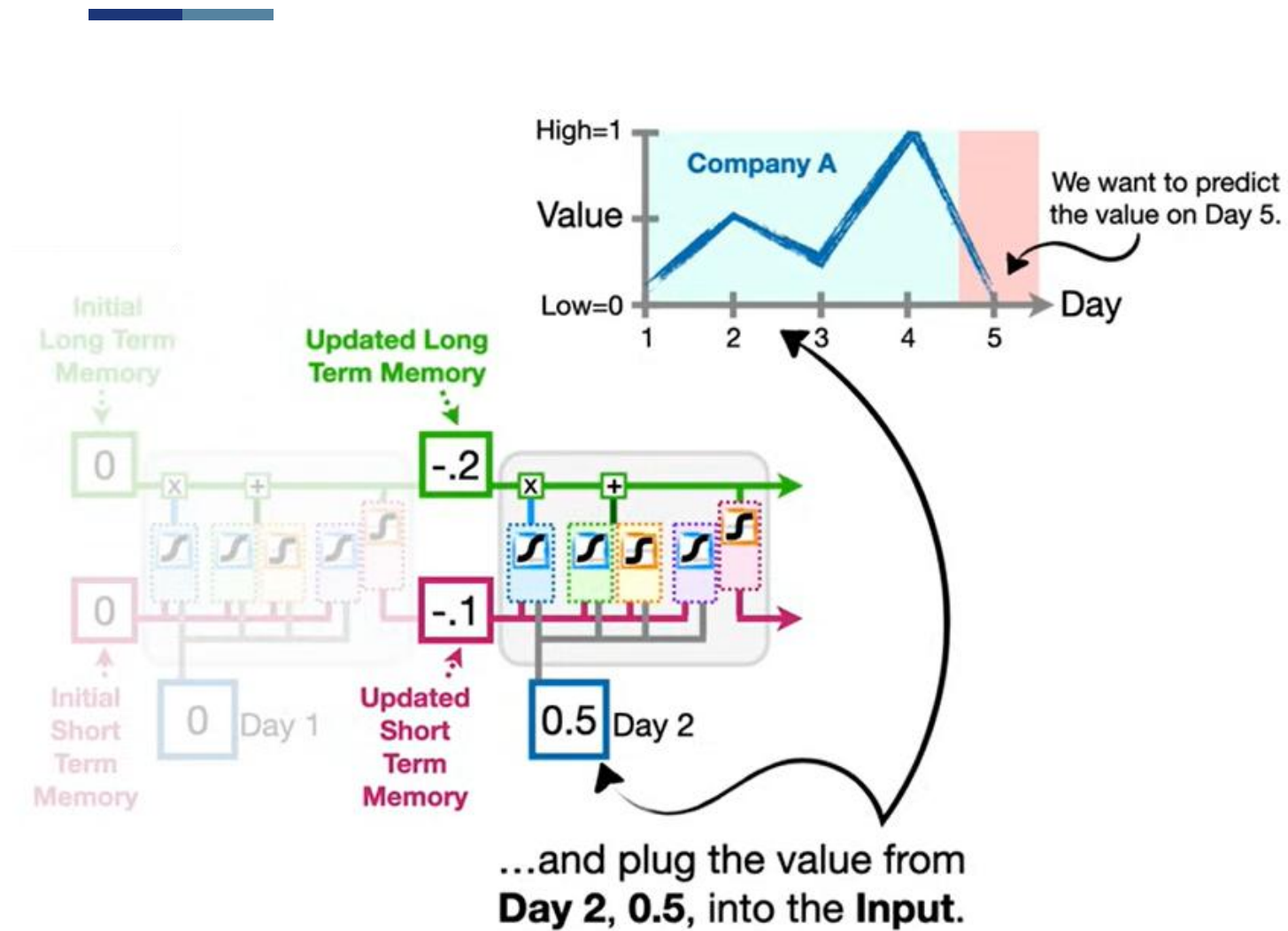


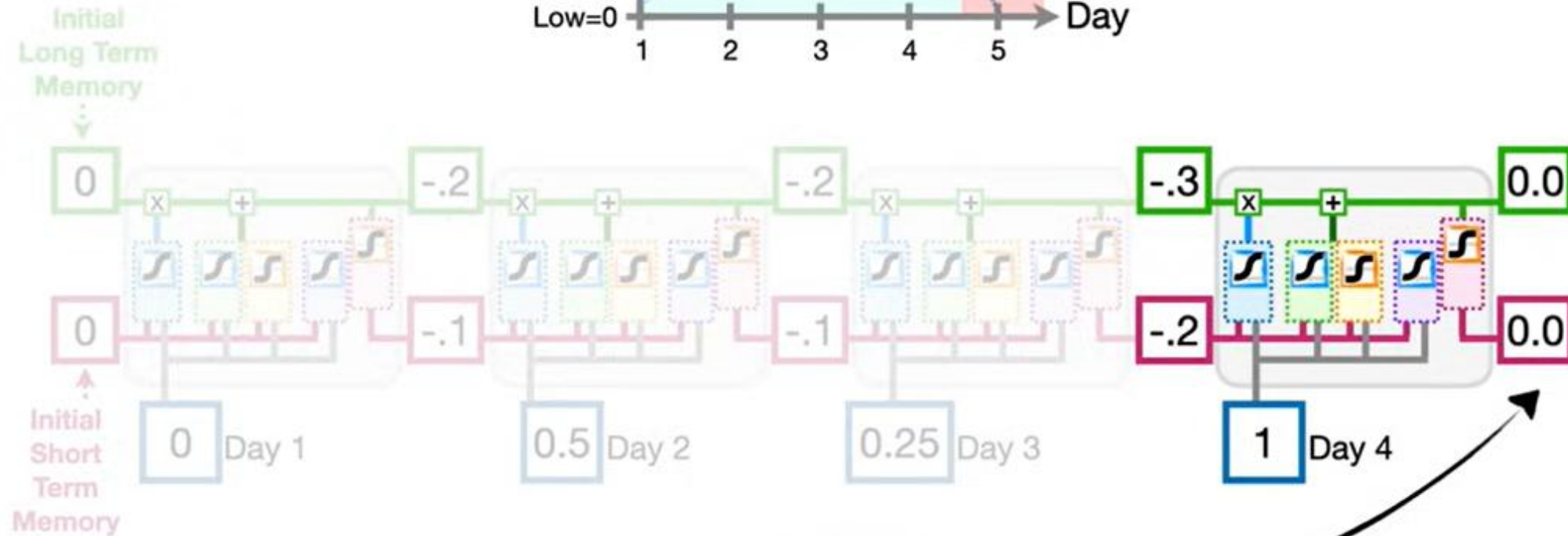
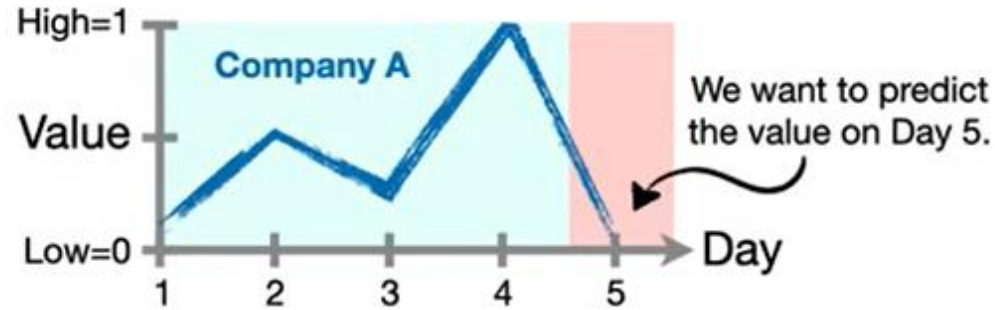
Now, if we want to sequentially run **Company A**'s values from **Days 1** through **4** through this **LSTM**...



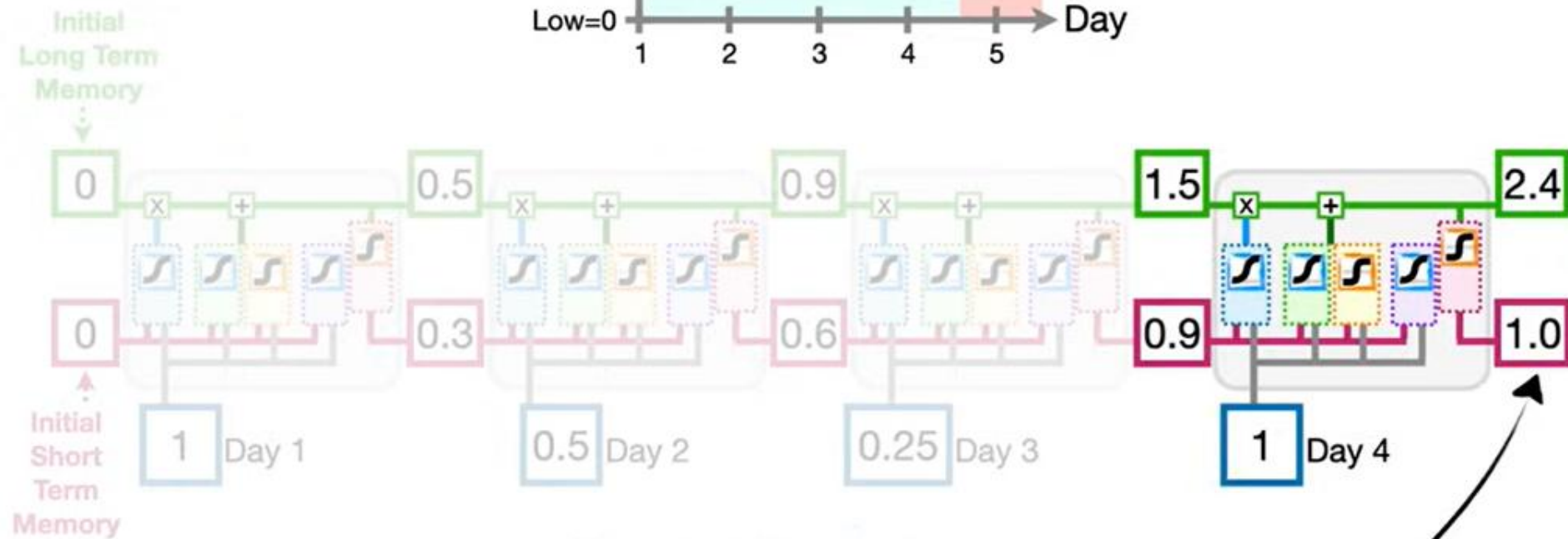
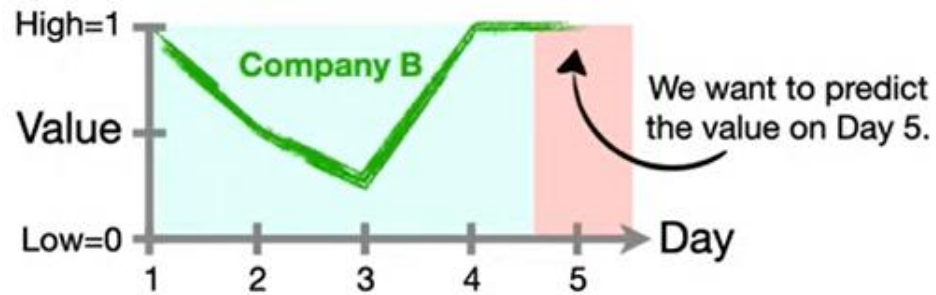


...and **-0.1** (rounded) for the updated **Short-Term Memory**.





And the final **Short-Term Memory**, **0.0**, is the **Output** from the unrolled **LSTM**.



...and the final **Short-Term Memory**, 1.0, is the **Output** from the **unrolled LSTM**.